

Worksheet 7

Shunt and series motor basics

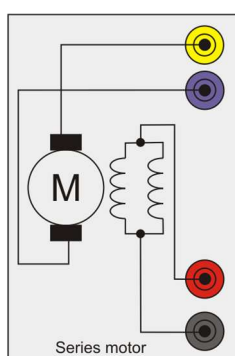
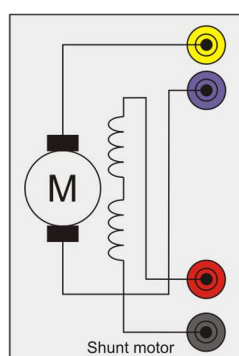
Modern electrical machines



Wound stator DC motors are often configured as either **shunt** wound or **series** wound.

The separate connections for the field windings of the stator and rotor / armature allow us to control the current in both parts of the motor separately, in order to examine the motor's behaviour.

The photograph shows an old shunt wound motor.



Red and black terminals are for the armature winding, yellow and purple terminals for the rotor windings.

Over to you:

- 1) With both the series and shunt motors disconnected from the test rig, use a digital multimeter to measure the resistance of the field windings on both the shunt and series motors.
- 2) Use the Ohm's law formula, $I = V/R$, to calculate the theoretical field currents when the field windings are connected to a 24V DC supply.
- 3) Repeat the resistance measurement for the armature windings
- 4) Record your results in the table opposite.
- 5) In practice, you get different values of current which we discuss below.

Armature windings (field coils)

Motor	Resistance in Ω	Current in A
Shunt		
Series		

Rotor windings

Motor	Resistance in Ω	Current in A
Shunt		
Series		

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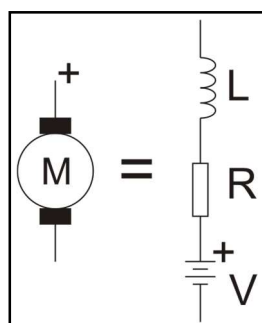
So what?

Electrical machines are made up of multiple windings. When DC flows through them, their electrical characteristics can be predicted using the Ohm's law formula. However, once the electrical machine is running the rotating magnetic fields in the machine alter their behaviour drastically.

The equivalent circuit of the machine then becomes more complicated. In particular, the rotating magnetic components in the machine induce an electromagnetic force (e.m.f.) which acts to oppose the flow of current in the windings. You can see the equivalent circuit below.

Understanding how to calculate this opposing e.m.f. is outside the scope of this learning package - but you need to be aware that it exists.

The reverse e.m.f. increases as speed increases. When a machine stalls, therefore, the reverse e.m.f. is reduced. The current flowing then increases significantly, increasing the risk of burning out the motor .



Equivalent circuit of a simple DC machine.

Over to you:

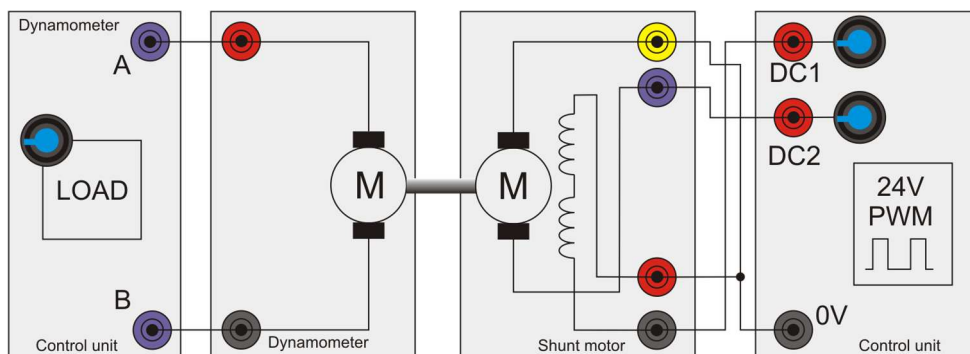
- 1) Connect the multimeter to the rotor winding of the Shunt motor on the ohmmeter setting.
- 2) Slowly turn the motor shaft. Explain what you see on the meter.

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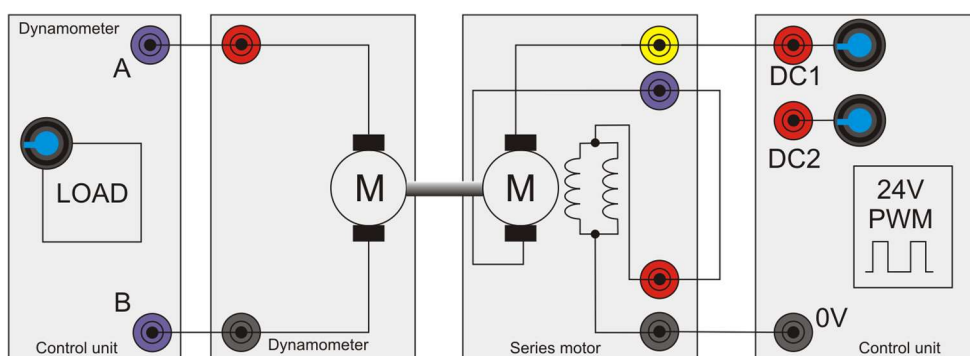
Shunt and series motor basics

Modern electrical machines

Shunt wound



Series wound



Note: the circuits are similar but not quite the same.

With all wound-stator DC motors, you swap over the connections to **either** the stator **or** rotor windings to make the motor rotate in the opposite direction.

Reversing the voltage to both sets of windings keeps the motor rotating in the same direction, as both fields are reversed.

Over to you:

- 1) Set up the **shunt** motor as in the top diagram.
- 2) On your PC, load the 'Open Control_DC.bat' program. Set the output on DC1 (the field coils) to 100% or 24V. Set the output on DC2 (the rotor) to around 70% and press 'RUN'.
- 3) Increase DC2 in steps of 10% up to 100%. In the table on the next page, record the voltages and currents at each stage.
- 4) Now, change to the **series** motor and wire it up as in the lower diagram.

5) Set the output on DC1 to 40% and press RUN.

6) Increase DC1 in steps of 5%. Again, record the output voltages and currents each time. The currents eventually gets so high that the control unit cuts out.

So what?

This experiment shows that the speeds of the series and shunt motors are very different for similar voltage inputs .

The control unit cuts out at a certain speed (current) for the series motor. The current is limited to protect the motor and the control unit.

The control unit delivers an output power up to 100W. It cuts out when this is exceeded and shows an 'Err' message on the front of the unit. The software application stops.

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Shunt motor

DC1	%	DC2	%	I1 in A	I2 in A	Speed in RPM
100		70				
100		80				
100		90				
100		100				

Series motor

DC1	%	I1 in A	I2 in A	Speed in RPM
40				
45				
50				
55				

Indicate clearly the point at which the control unit cuts out.

So what?

The speed of the Series motor is very high. The Control unit is designed to cut out at speeds above 3100 RPM. Above this the Dynamometer produces voltages that can be dangerous.